IT3091 Machine Learning – Group Assignment Descriptor

guided reasoning, business lenses, evidence-based ML decisions

1. Overview: two assignment tracks

This assignment gives each group a platform to practise the capabilities required in the AI era: problem framing, critical thinking, data/statistical reasoning, ML implementation, responsible AI judgement, communication, collaboration, and learning how to learn. The dataset or business scenario is the starting point; the technical pathway must be reasoned and justified by the group.

Track	What is provided / expected	What students must decide
Guided Data Track	The module team provides a business scenario and dataset/source. Students work within a controlled domain.	Primary lens, optional secondary lens, exact ML/data-mining task, target/output, preprocessing, models/methods, evaluation metric, and recommendation.
Industry Explorer Track	Students identify an approved real-world business problem and obtain/collect a suitable dataset. Prior approval is required.	Same technical requirements as Guided Data Track plus evidence of real-world engagement, ethical handling, and practical business value. Eligible for up to 10 bonus marks.

2. Business lenses

Each group must choose one primary decision lens. One secondary lens is allowed only when it directly strengthens the primary decision. A secondary lens is optional; doing more work without coherence will not earn higher marks. Marks are awarded for defensible reasoning, not for attempting many disconnected analyses.

3. Guided Data Track: business scenarios and possible lenses

Code (Based on last digit of your group)	Domain	Dataset Link	Business Scenario	Possible Decision Lenses	Student Responsibility
0	Retail & E-commerce	Online Retail	An online retailer wants to improve sales, customer retention, and product strategy using historical transaction data.	Customer segmentation; product association/bundling; customer value/retention	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
1	Healthcare	Diabetes 130-US Hospitals	A hospital wants to use historical patient records to support planning and reduce avoidable patient risk.	Readmission risk; high-risk patient profiles; patient grouping/resource planning	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
2	Banking & Finance	Bank Marketing	A bank wants to improve the effectiveness of a marketing campaign and use customer information more intelligently.	Campaign response prediction; customer segment prioritisation; marketing profile analysis	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
3	Education	Student Performance	An academic institution wants to identify patterns that can support student performance and early intervention.	Performance prediction; at-risk student identification; intervention group profiling	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
4	Agriculture	Rice Cammeo and Osmancik	An agricultural organisation wants to improve crop/product classification and quality-related decisions using measured features.	Variety classification; feature-importance analysis; quality-control decision support	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
5	Logistics & Supply Chain	DataCo Smart Supply Chain	A logistics company wants to improve delivery reliability and reduce operational delays.	Late-delivery risk; delay pattern analysis; shipment priority/risk profiling	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
6	Tourism & Hospitality	Hotel Booking Demand	A hotel chain wants to improve booking reliability, revenue planning, and customer management.	Cancellation risk; booking/customer segmentation; operational planning support	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
7	Telecommunications	Telco Customer Churn	A telecom provider wants to reduce customer loss and improve retention decisions.	Churn risk; high-risk customer profiling; retention segment analysis	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
8	Real Estate	Ames Housing / House Prices	A real estate company wants to understand property value factors and support better pricing decisions.	Price prediction; valuation feature analysis; property segment analysis	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.
9	Sports Analytics	Football-Data	A sports organisation wants to use historical match data for performance analysis and strategic decision-making.	Match outcome analysis; team performance patterns; home advantage/form analysis	Choose and justify the lens, formulate the exact task/output, design the workflow, compare methods, evaluate, and recommend.

4. Core evidence required from every group

Problem framing canvas	Workflow diagram	Decision log	EDA insight log	Preprocessing/feature log	Model/method comparison	Recommendation + limitations

5. Main marking rubric: what this assignment encourages

Criterion	Marks	High-mark evidence	Fink alignment	AI-era skill
Business problem framing and lens/task formulation	5	Clear stakeholder, decision need, primary lens, optional secondary lens, unit of analysis, exact task/output, and rationale.	Foundational Knowledge; Integration	Problem framing; critical thinking
Workflow diagram and decision log	10	A coherent workflow from business problem to recommendation; important decisions recorded with options, reasons, and evidence.	Integration; Learning How to Learn	Reasoning; communication
Data understanding, EDA, and data quality reasoning	10	Data dictionary, row meaning, variable types, patterns, data issues, and explanation of how issues affect the task.	Foundational Knowledge; Application	Data literacy; statistical reasoning
Preprocessing and feature-engineering decisions	15	Justified handling of missing values, outliers, duplicates, categories, scaling, imbalance, leakage, and non-tabular features where relevant.	Application	ML implementation; data reasoning
Model/data-mining strategy and comparison	20	Sensible baseline and at least three alternatives; choice of methods justified by task, data, interpretability, and constraints.	Application; Integration	ML implementation; problem solving
Evaluation, validation, and critical judgement	20	Appropriate metrics, train/test or cross-validation logic, leakage prevention, honest interpretation, and comparison beyond accuracy.	Application; Integration	Statistical reasoning; responsible AI
Recommendation, limitations, and stakeholder value	10	Evidence-based practical recommendation, clear limitations, risks, and stakeholder value.	Integration; Human Dimension; Caring	Communication; human-centred AI
Reproducibility, documentation, and AI-use transparency	10	Notebook runs clearly, outputs match the report, dataset link/fingerprint is present, and AI-use is honestly declared.	Learning How to Learn	Professional practice; responsible AI

6. Submission requirements

Stage	Group/Individual	Required submission
Initial	Group	Initial Submission. Include track, lenses with reasons, proposed task/output, workflow, and core responsibility of each member.
Final	Group	3 minute YouTube demo video. Final report with final notebook/code, dataset/data dictionary, decision logs model/method comparison.
Final	Individual	One A4 Personal Learning Journey report. This will support assessment of individual contribution,

7. Industry Explorer Track bonus rubric

Industry Explorer groups follow the same 100-mark core rubric. They may receive up to 10 additional bonus marks only after approval and evidence review. The bonus is not awarded simply for finding a dataset online.

Bonus criterion	Marks	Evidence required
Real stakeholder or business context identified	2	Named stakeholder/context, decision need, and realistic value.
Problem scope approved and suitable for ML/data mining	2	Problem approved early; manageable within seven weeks; not a full software/deep learning project.
Dataset is original, local, stakeholder-provided, or meaningfully real-world	2	Data source is verifiable; dataset fits the selected decision lens.
Data collection method, data dictionary, and privacy handling documented	2	Permission/anonymisation where relevant; no unsafe personal/sensitive data handling.
Final recommendation has practical business value	2	Recommendation is evidence-based and meaningful for the stakeholder.

8. Final message

This assignment is not about following a fixed recipe. The highest marks will go to groups that make the most defensible machine learning/data mining decisions: clear problem framing, evidence-based data reasoning, justified preprocessing, suitable model/method comparison, honest evaluation, practical recommendation, responsible AI judgement, and clear individual learning.